

Joan Alcaide-Núñez

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[Website](#) — [LinkedIn](#) — [GitHub](#)

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Research Experience

GRB Cosmography with Empirical Energy Relations

June–July 2025

Osservatorio Astronomico di Brera (OAB-INAF), Italy

Applied GRB empirical relations to constrain cosmological parameters via χ^2 minimization. Combined GRB results with binary neutron star simulated populations from Einstein Telescope expectations. Received mentoring on relativistic effects and theoretical foundations of GRBs. Participated in weekly journal clubs and research seminars. [code repository](#) Supervisors: Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia

CAPIBARA Collaboration

2024–Present

Principal Investigator, Founder

Student research collaboration developing instrumentation and mission concept for high-energy astronomy CubeSat mission. [project website](#)

Student Internship at German Space Agency

October 2024

Institute for Remote Sensing (DLR-IMF), Germany

Worked with experimental methods and photogrammetry departments on hyperspectral imaging and Landsat thermal data analysis. Conducted field measurements and visited German Space Operations Center (GSOC).

Type Ia Supernovae Distance Measurements

June–July 2024

Institute of Space Sciences (ICE-CSIC-IEEC), Spain

Computed Type Ia SNe distances using infrared and optical photometry from ESO, ATLAS, and ZTF. Fitted cosmological parameters including the Hubble constant. Developed proficiency in astronomical data analysis. Participated in journal clubs and group meetings. Supervisor: Dr. Lluís Galbany

Cepheid Variable Stars Research

August–December 2023

Youth & Science Program (year 1)

Computed Hubble constant using Cepheid data from NASA-IPAC NED database and Konkoly observatory extragalactic cepheid database, recovering $H_0 = 70.26 \pm 7.1 \text{ kms}^{-1}\text{Mpc}^{-1}$. First experience with astronomical data analysis, paper reading, and scientific writing. [paper report](#) Supervisors: Dr. Ignasi Pérez-Ràfols, Dr. Laia Casamiquela

Education

Ludwig-Maximilians-Universität (LMU) Munich

September 2025 – Present

B.Sc. in Physics

German School of Barcelona

September 2017 - May 2025

Dual High School Diploma (Germany + Spain), GPA: 1.0/1.0

Additional Training:

- *Gravity and Black Holes* — Perimeter Institute (GoPhysics!), 2025

- Mathematics, Statistics, and Data Science — Universitat Autònoma de Barcelona (UAB), 2022
- Algorithms and Programming (C++) — Universitat Politècnica de Catalunya (UPC), 2022
- League of Codes (C++) — Harbour Space, 2022-2023

Awards

Youth & Science fellowship: 3-year funded [fellowship](#) of research experiences for high school students. Prestigious scholarship that allowed me to have early first-hand experience in research stays internationally.

German Physical Society (DPG): Physics Distinction for Academic Achievement in high school.

International Astronomy & Astrophysics Competition (IAAC): Silver Honour and National Award for Spain (2025), Silver Honour (2024), Bronze Honour and National Award Spain (2023). In 2025, top 7% students ($n > 12000$).

Jugend Forscht: First Prize Iberia 2025 ("*Cosmological distance measurements with Type Ia Supernovae*"), First Prize and Special Award in Scientific Photography NRW 2025; First Prize Iberia 2024 ("*Recomputing the Hubble constant*"), DLR Internship Award NRW 2024.

Talks

The CRD Instrument and the COSMOS program April 2026
5th Symposium on Space Educational Activities (SSEA)
 Presented ongoing work by the CAPIBARA Collaboration.

De l'aula a l'espai: joves, ions i fotons per entendre l'univers d'altres energies (Contributed) March 2025
CosmoXarxa, CosmoCaixa Barcelona's Science Museum
 CosmoXarxa is an initiative created by and for Explainers of the science museum (outreach volunteers). I presented the CAPIBARA project, its objectives and status, both technical, scientific and educational, inviting the public to join the efforts. [slides](#)

Introduction to Astrophysics (Invited) May/June 2024
German School of Barcelona
 Double session (3h) about a naive introduction to astronomy and astrophysics for 11th graders, covered some cosmology (Big Bang theory), star evolution, galaxy morphology and massive bodies (neutron stars and black holes as extreme cases of star evolution). [slides](#)

L'Exploració Espacial (Invited) October 2023
Escola Canigó
 Presentation to pre-school students about human space travel and solar system exploration. Focused on hands-on learning with models and didactic, entertaining content. [slides](#)

Explainers at CosmoCaixa Museum 2022-2023
 Volunteered as explainer at Barcelona's Science Museum explaining demonstrations to visitors on weekends. Participated in events such as Explainers during Winter, Explainers at Barcelona's City Science Fair and CosmoXarxa.

Skills

Python: numpy, scipy, pandas, matplotlib, astropy

Tools: Git, GitHub, L^AT_EX, Markdown, Visual Studio, NeoVim